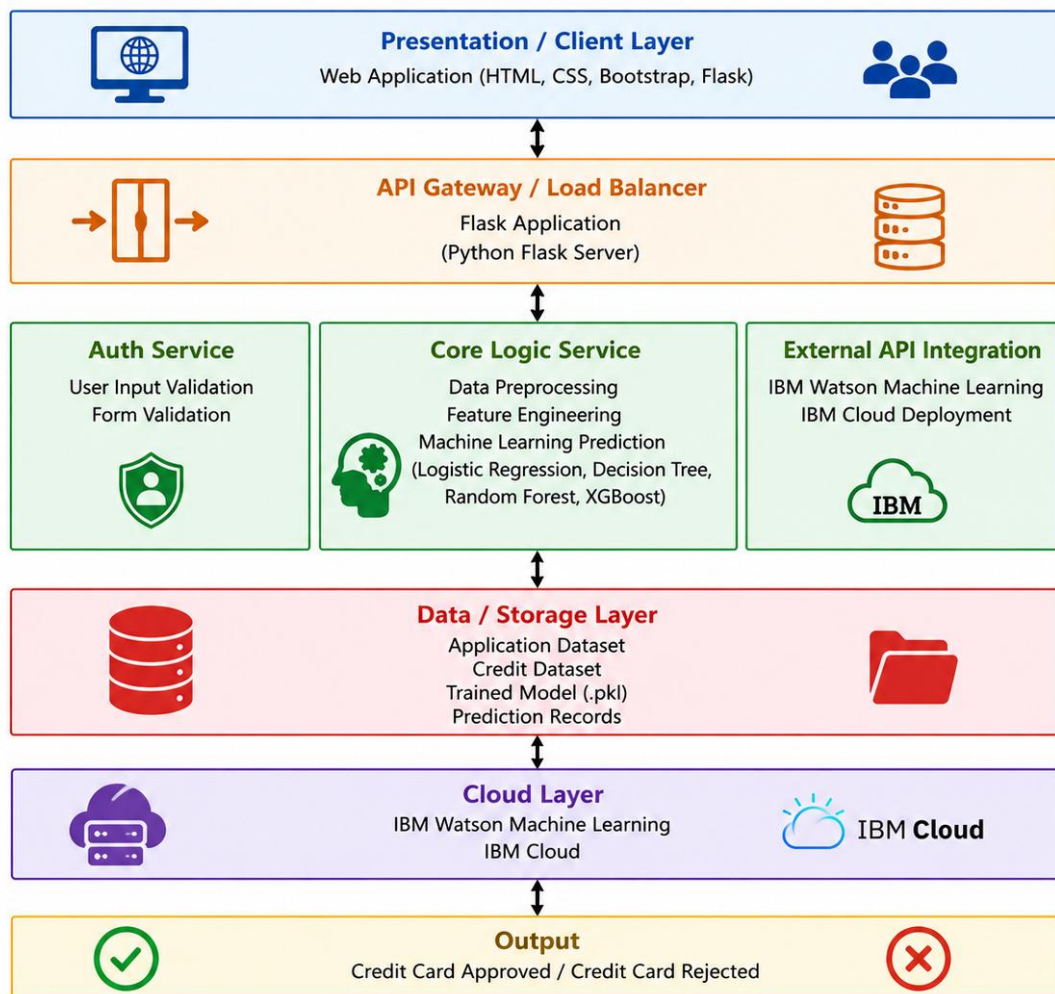


Solution Architecture

Date	1 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Credit Card Approval Prediction System follows a layered architecture. The user submits applicant information through the Flask web interface. The Flask backend validates the input and sends it to the trained machine learning model. After preprocessing and prediction, the system returns either an **Approved** or **Rejected** result. The trained model and datasets are maintained in the data layer, while IBM Cloud provides deployment and scalability for real-time predictions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where applicants or bank analysts enter financial details and view the credit card approval prediction results.	HTML, CSS, Bootstrap, Flask Templates
API Gateway	Receives user requests, validates input data, forwards the request to the machine learning model, and returns the prediction result to the presentation layer.	Python, Flask
Microservices	Performs data preprocessing, feature engineering, loads the trained model, executes prediction using the best-performing classifier, and processes the response.	Scikit-learn, XGBoost, Pandas, NumPy, Pickle
Database	Stores applicant datasets, credit records, trained machine learning model, and prediction-related data required by the application.	CSV Files, Pickle (.pkl), IBM Cloud Storage (for deployment)